

Decision-focused fine-tuning of time series foundation models for dispatchable feeder optimization

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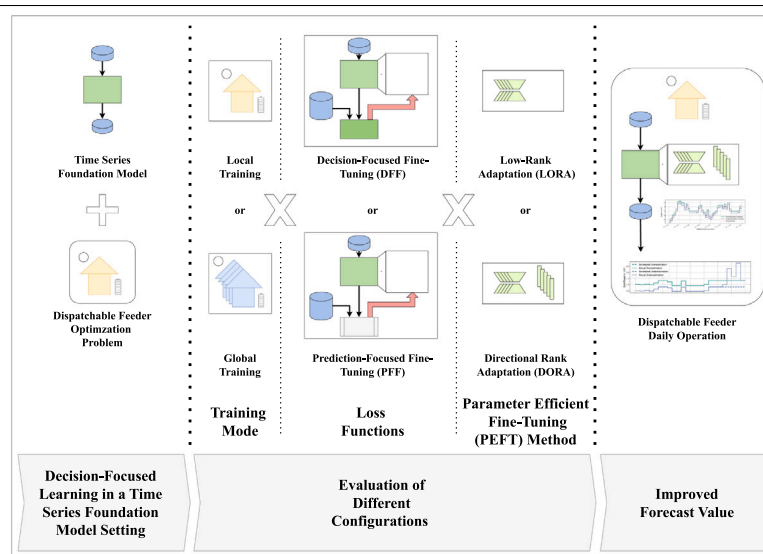
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HIGHLIGHTS

- Combines decision-focused learning with Parameter Efficient Fine-Tuning (PEFT).
- Integrates decision-focused learning into fine-tuned time series foundation models.
- Combines a fine-tuned time series foundation model with an energy optimization use case.

GRAPHICAL ABSTRACT



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ABSTRACT

Time series foundation models provide a universal solution for generating forecasts to support optimization problems in energy systems. Those foundation models are typically trained in a prediction-focused manner to maximize forecast quality. In contrast, decision-focused learning directly improves the resulting value of the forecast in downstream optimization rather than merely maximizing forecasting quality. The practical integration of forecast values into forecasting models is challenging, particularly when addressing complex applications with diverse instances, such as buildings. This becomes even more complicated when

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instances possess specific characteristics that require instance-specific, tailored predictions to increase the forecast value. To tackle this challenge, we use decision-focused fine-tuning within time series foundation models to offer a scalable and efficient solution for decision-focused learning applied to the dispatchable feeder optimization problem. To obtain more robust predictions for scarce building data, we use Moirai as a state-of-the-art foundation model, which offers robust and generalized results with few-shot parameter-efficient fine-tuning. Comparing the decision-focused fine-tuned Moirai with a state-of-the-art classical prediction-focused fine-tuning Moirai, we observe an improvement of 9.45% in Average Daily Total Costs.

1. Introduction

Decision-making processes play a critical role across various domains. In some cases, these processes are formulated as end-to-end optimization problems within an environment — such as in reinforcement learning — where decision-making is directly optimized in-the-loop with respect to the environment. Examples include battery arbitrage [1], dynamic combustion optimization [2,3], and portfolio optimization [4]. In these settings, explicitly adjusting the forecast isn't always necessary because the decision-maker learns from interacting with the environment.

However, many real-world applications follow a classical subsequent optimization paradigm, which enables a clear mathematical formulation of the problem and facilitates the computation of an optimal solution with respect to the forecast. In this approach, forecasts are first generated and then used as inputs to downstream optimization problems. Applications of this paradigm span various domains, including inventory optimization [5], portfolio optimization [6], energy system optimization [7], building-level schedule optimization [8], and building-level probabilistic schedule optimization [9]. However, even though the forecasts are mainly generated as inputs for downstream optimizations, they are often evaluated and trained using forecast quality metrics, such as Mean Square Error (MSE) or Mean Absolute Error (MAE). Those forecast quality metrics do not necessarily reflect the actual value for the downstream optimization problem. Literature shows the discrepancy of forecast quality and value within various applications, e.g., in building asset optimization [10], inventory planning [11], photovoltaic storage optimization [12], or in dispatch schedule optimization [13].

To address the challenge and divergence in objective, decision-focused learning aims to optimize the forecasting model to directly improve forecast value rather than merely maximizing forecast quality [14]. With that, decision-focused learning contrasts with the training paradigm of prediction-focused learning, which focuses on the forecast value in the model learning process. However, the practical integration of forecast value into learning models remains challenging, particularly when considering large-scale applications carried out on various instances, which need tailored forecasts related to the instance-specific properties. An “instance” refers to a specific entity within a system that has unique properties that affect the optimization process. For example, in a residential battery optimization problem, each individual building is considered an instance with varying characteristics, such as its Photovoltaic (PV) installation and consumption patterns, which influence the optimization outcomes [13]. [15] proposes a scalable solution across instances of their optimization system by proposing a surrogate neural network within a decision-focused retraining process to generate suitable forecasts.

To optimize at the residual level, accurate and suitable forecasts at the household level are required in the energy sector. Recently, foundation models have gained traction in the load forecasting community [16], offering a robust solution to address data scarcity [17] while achieving state-of-the-art performance [16]. These models serve as an excellent baseline for fine-tuning, having been trained prediction-focused across a diverse range of time series tasks. However, their out-of-the-box performance may not capture the unique instance-specific fluctuations that play an important role in household-level forecasts.

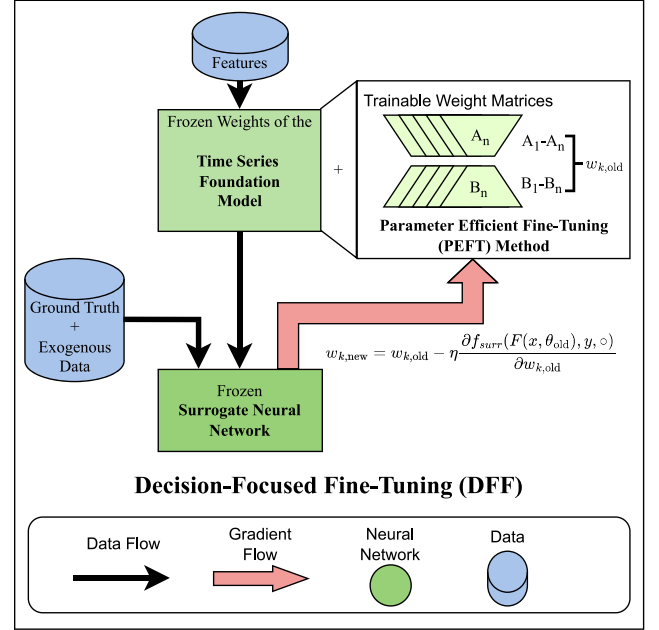


Fig. 1. An overview of the decision-focused fine-tuning process which combines surrogate neural networks with Parameter Efficient Fine-Tuning (PEFT) methods. It employs the technique from [15] using a Surrogate Neural Network and its gradient to influence the weights of the PEFT with respect to the downstream optimization.

Therefore, this paper makes a threefold contribution at the intersection of decision-focused learning, foundation models, and energy forecasting. By integrating these domains, we address the need for decision-focused forecasting while leveraging the strengths of time series foundation models in an energy optimization use case context. The three contributions are:

First, this work introduces a novel approach that combines decision-focused learning with fine-tuning techniques applied to pre-trained models. Second, it is the first study to integrate decision-focused elements into the fine-tuning of time series foundation models. The resulting method, Decision-Focused Fine-Tuning (DFF), illustrated in Fig. 1, builds on the retraining methodology proposed in [15], providing a scalable and efficient solution for energy systems with numerous instances. Third, within the energy community, this work is the first to connect time series foundation models with decision-focused learning within an energy-related optimization use case. Furthermore, it demonstrates a global model that requires training only once, providing an efficient and scalable solution with improved performance compared to purely prediction-focused global models.

Importantly, this work focuses on the inherent evaluation of the time series foundation model Moirai [18] within the DFF setting. It contrasts this approach with a prediction-focused learning paradigm by evaluating both forecast value and traditional prediction metrics. Moirai was not chosen for benchmarking, but because it represents a promising class of general-purpose forecasting time series foundation models that can be flexibly applied across datasets and use cases. Its strong out-of-the-box performance makes it a suitable starting point for fine-tuning, particularly in the direction of maximizing forecast value.

This work deliberately omits comparisons with fully trained models such as DeepAR or PatchTST, as recent results indicate similar or better predictive performance in a more computationally efficient manner with zero-shot forecasts from modern foundation models [19]. Furthermore, since we aim to present a broad solution applicable for multiple tasks, we do not include highly specialized domain models tailored to narrow tasks, such as NOx emission prediction [20], entropy-based wavelet packet ensemble model electrical load forecasting [21], physics-inspired neural network wind power forecasting [22] or other domain-specific architectures.

The remainder of this paper is structured as follows. First, a brief overview of related work is provided in Section 2, covering decision-focused learning, foundation models, and fine-tuning methods. It is followed by a detailed description of the proposed decision-focused fine-tuning methodology in Section 3. An experiment applying DFF on an exemplary optimization problem and a study of data usage in different fine-tuning modes and two different fine-tuning methods are outlined in Section 4 and Section 5. The results are shown in Section 6 and discussed in Section 7. Finally, Section 8 concludes the findings of this research.

2. Related work

This section is fourfold. First, it introduces foundation models. Second, it introduces how those models are adapted and fine-tuned using Parameter Efficient Fine-Tuning (PEFT) approaches. Afterward, it introduces the paradigm of learning to decision-focused learning. It concludes by pointing out the contribution.

2.1. Foundation models

Foundation models are used in various domains, such as computer vision [23], e.g., SAM 2, [24] or Florence-2 [25], natural language processing [26], graph learning [27] and time series forecasting [28]. They are known for their broad generalizability. Currently, there are various foundation models for time series, e.g., Chronos [29], TimesFM [30], MOMENT [31], Lag-Llama [32], Moirai [18] or TimeGPT-1 [33]. Time-series foundation models are useful, especially as they can be used zero-shot without training. Information about such models' technical background and taxonomy is listed in [28]. A detailed comparison and benchmarking of different data usage in zero-shot and few-shot forecasting can be found in [34].

In the literature of the energy domain, time series foundation models are partly evaluated in load forecasting. In short-term load forecasting, the foundation model Chronos [29] achieved zero-shot performance comparable to a specially fine-tuned transformer model in data from the BuildingsBench platform [16]. [35] showed that time series foundation models are capable of delivering competitive forecast performance compared to trained-from-scratch transformer models. Foundation models can be fine-tuned with available data to increase their task-specific performance. For example, the authors of [17] fine-tuned the foundation model TimeGPT-1 [33] for load forecasting in a data-scarce environment.

2.2. Parameter efficient fine-tuning

With the rise of foundation models, methods for Parameter Efficient Fine-Tuning (PEFT) are emerging. PEFT methods are especially useful for fine-tuning large models efficiently, as they only adjust a few parameters, leading to cost-reduced training, as the zero-shot performance is improvable by adapting available data. Two reviews are given in [36,37]. In this paper, we focus on two exemplary reparameterized fine-tuning methods, adapting selected weights of the neural network architecture:

LoRA. (Low-Rank Adaptation) [38] uses two low-rank matrices, where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and the rank $r \ll \min(d, k)$, with the original frozen weight matrix $W_0 \in \mathbb{R}^{d \times k}$. The resulting matrix W' can be derived through

$$W' = W_0 + \Delta W = W_0 + BA. \quad (1)$$

Therefore, weight updates in LoRA result from adapting the trainable weights low-rank matrices B and A .

DoRA. (Directional Rank Adaptation) [39] builds on LoRA by adapting the direction of pre-trained weight matrices. Similarly, as in LoRA, weight updates are achieved using low-rank matrices B and A . DoRA extends this by introducing a trainable vector m , initialized with $\|W_0\|_c \in \mathbb{R}^{1 \times k}$ and $\|\cdot\|_c$ being the vector-wise norm of a matrix across each column, scaling the direction of weight update. It keeps the pre-trained weight W_0 fixed and modifies by updating the weights in BA . Within DoRA the resulting weight matrices can be written as

$$W' = m \frac{W_0 + BA}{\|W_0 + BA\|_c}. \quad (2)$$

Therefore, weight updates in DoRA result from adapting the trainable weights low-rank matrices B and A and the vector m .

2.3. Decision-focused learning

A review of decision-focused learning is given in [14]. The main goal of decision-focused learning is to optimize forecasts concerning the subsequent downstream optimization. Literature uses the concept of surrogate losses to optimize the upstream forecaster within different use cases, e.g., of energy [15,40], health supply chain [41], versions of Cubic Top-K [42], portfolio optimization [43], quadratic and broader nonlinear portfolio selection [44], and web advertisement [43]. These surrogate losses form useful gradients concerning the optimization problems' decision costs and, therefore, can be used to learn the needs of downstream optimization. This can be accomplished by learning a parametrization of a convex loss function, i.e., a parameterized Mahalanobis loss [45], or the local optimized decision loss family as in [43] or efficient global losses as in [42]. Further possibilities are through an optimal piecewise linear approximation method and the Huber norm embedding technique [40], by reweighting the MSE [41,46], or by learning a landscape surrogate loss [44]. Decision-focused retraining [15] learns a surrogate neural network, which serves as surrogate loss and retrains the forecaster in a subsequent training process after the initial training.

2.4. Own contribution

The contribution combines all the aspects given in this section by making a threefold contribution.

1. To the best of the authors' knowledge, this work presents the first approach that combines decision-focused learning with PEFT to efficiently fine-tune pre-trained models.
2. This is the first study to integrate decision-focused learning into fine-tuning time series foundation models to decrease downstream decision costs.
3. Within the energy community, this work is the first to connect a time series foundation model with an exemplary energy-related optimization problem and apply decision-focused learning.

3. Decision-focused fine-tuning

This section combines aspects of decision-focused learning using surrogate networks from [15] and fine-tuning of foundation models using PEFT methods. It introduces Decision-Focused Fine-Tuning (DFF) (visualized in Fig. 1), derived from the main objective of decision-focused learning. The DFF approach is applied on a pre-trained time series foundation model, similar to [15].

Decision-focused learning aims to maximize the forecast value for downstream optimization. Specifically, we define the value function $V(F(x, \theta), y)$ that returns the forecast value with $F(x, \theta)$ being a forecaster F of y , with features x , parameterized by θ .¹ Consequently, the primary objective of decision-focused learning is to find the best parametrization of θ , i.e.,

$$\operatorname{argmin}_{\theta} (V(F(x, \theta), y)). \quad (3)$$

To minimize the function $V(F(x, \theta), y)$, decision-focused learning methods try to determine a gradient concerning the forecast value,

$$\nabla V(F(x, \theta), y) = \frac{\partial V(F(x, \theta), y)}{\partial \theta}. \quad (4)$$

The value function $V(F(x, \theta), y)$ is not differentiable as it needs the forecast applied within the optimization problem. In contrast, the value function can be modeled by a neural network, which we call a surrogate neural network and is thus differentiable by design as in [15]. This function can be used to replace $V(F(x, \theta), y)$ by $f_{\text{sur}}(F(x, \theta), y, \circ)$ with $\circ$ being exogenous factors influencing the outcome of the optimization problem. Therefore, we can write the above-defined gradient as

$$\nabla V(F(x, \theta), y) = \frac{\partial V(F(x, \theta), y)}{\partial \theta} = \frac{\partial f_{\text{sur}}(F(x, \theta), y, \circ)}{\partial \theta}. \quad (5)$$

To adapt time series foundation models, we use a PEFT method, meaning that instead of adapting the original weight matrix of the foundation model, only the weights of the PEFT method adapter are trained. Using backpropagation, the trainable weights w_k of the PEFT adapter are fitted to minimize the value function $V(F(x, \theta), y)$. I.e., using DoRA, the weights of the low-rank matrices B and A and the vector m (see) are updated. The updated weights $w_{k,\text{new}}$ can be described as

$$w_{k,\text{new}} = w_{k,\text{old}} - \eta \frac{\partial f_{\text{sur}}(F(x, \theta_{\text{old}}), y, \circ)}{\partial w_{k,\text{old}}}, \quad (6)$$

with the learning rate $\eta > 0$. In the exemplary case of DoRA, the weights in B , A , and m together with the not updated weights from the frozen weight matrix W_0 form the resulting parameters θ of the forecaster, see Eq. (2). This leads to weights fitted to the forecast value and not to forecast quality. The surrogate neural network can also be viewed as a surrogate loss function. This serves as an analogy to a loss function of prediction-focused learning, but in contrast, this function is aligned with the decision cost in the downstream optimization.

4. Application of decision-focused fine-tuning

This section investigates the degree to which DFF can increase the forecast value when using a forecast from a foundation model in a downstream optimization.² To explore this, we focus on one exemplary optimization problem: the dispatchable feeder problem. This section mainly addresses the experimental setup, which is kept constant throughout. Therefore, this section first introduces the optimization problem briefly in Section 4.1. Afterward, the resulting forecast value from the optimization problem and the data used to realize the optimization problem are presented in Section 4.3. We then describe the applied models and their training in Section 4.4. In Section 4.5 the forecasting scenario and the fine-tuning setting are outlined before the benchmark methods are listed in Section 4.6.

4.1. Dispatchable feeder optimization problem

The dispatchable feeder optimization problem [8] is a hierarchical two-level non-convex optimization problem concerning the scheduling of a dispatchable feeder, i.e., a system consisting of residential power consumption, PV and a battery able to adhere to an optimized schedule. [8,47] define a dispatchable feeder as a system consisting of two main components: The first is the inflexible and variable energy prosumption of a residential building, defined as the residential load minus the power generated from the PV system. The second is a flexible and controllable residential battery system, subject to its physical constraints. The dispatchable feeder leverages the battery system to effectively manage the grid interaction, addressing the uncertainties associated with consumption and generation. The daily operation is structured on two levels. A day-ahead dispatch schedule (DS) is created in the first level based on prosumption forecasts. In the second level, the actual dispatch is determined by considering the real prosumption minimizing any deviation from the DS. The optimization problem is the same as in [15]. A more detailed description and all mathematical symbols are listed in Appendix A.1.

Forecast value of the dispatchable feeder optimization problem. We assess the forecast model's impact on the dispatchable feeder optimization problem using the Daily Total Costs metric established in [8]. Lower Daily Total Costs indicate higher forecast value.

The total costs encompass both the DS costs C_{DS} of the first-level optimization problem and the imbalance costs C_{Imb} incurred due to deviations from the DS during actual dispatch, handled in the second-level optimization. The DS costs promote self-consumption and peak shaving and are modeled by the cost function

$$C_{\text{DS}}(\bar{P}_g^+(k), \bar{P}_g^-(k)) = c_q^+ \cdot (\bar{P}_g^+(k) \cdot \Delta t)^2 + c_1^+ \cdot \bar{P}_g^+(k) \cdot \Delta t + c_q^- \cdot (\bar{P}_g^-(k) \cdot \Delta t)^2 + c_1^- \cdot \bar{P}_g^-(k) \cdot \Delta t, \quad (7)$$

with the positive DS $\bar{P}_g^+(k) \geq 0$, and the negative DS $\bar{P}_g^-(k) \leq 0$, the cost coefficients $c_q^+, c_1^+, c_q^-, c_1^- \in \mathbb{R}_{\geq 0}$ and the time interval Δt . The imbalance costs represent deviations between the scheduled and actual dispatch. With the deviation $\Delta P_g(k)$ the imbalance costs can be expressed as

$$C_{\text{Imb}}(\Delta P_g(k)) = c_q^A \cdot |\Delta P_g(k) \cdot \Delta t|^2 + c_1^A \cdot |\Delta P_g(k) \cdot \Delta t|, \quad (8)$$

with $c_q^A \in \mathbb{R}_{\geq 0}$ and $c_1^A \in \mathbb{R}_{\geq 0}$ being weighting coefficients.

The total costs are then calculated as:

$$C_T(\bar{P}_g^+(k), \bar{P}_g^-(k), \Delta P_g(k)) = C_{\text{DS}}(\bar{P}_g^+(k), \bar{P}_g^-(k)) + \alpha \cdot C_{\text{Imb}}(\Delta P_g(k)), \quad (9)$$

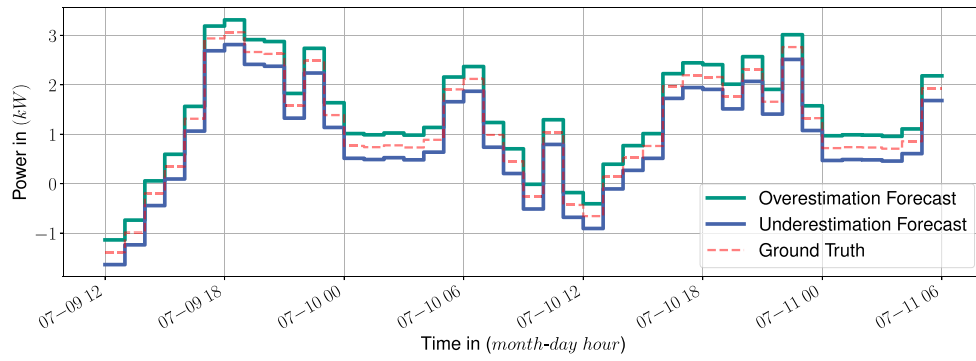
where $\alpha \in \mathbb{R}_{\geq 0}$ is a scaling factor for the costs of the imbalance. We select $\alpha = 10$ as in [48] to highly penalize the deviation in this scenario. Daily Total Costs are computed by summing the total costs over each day. For evaluation, the costs are averaged over all days and buildings under consideration, which we call Average Daily Total Costs, as in [15].

4.2. Case study: Forecast value for dispatchable feeder optimization

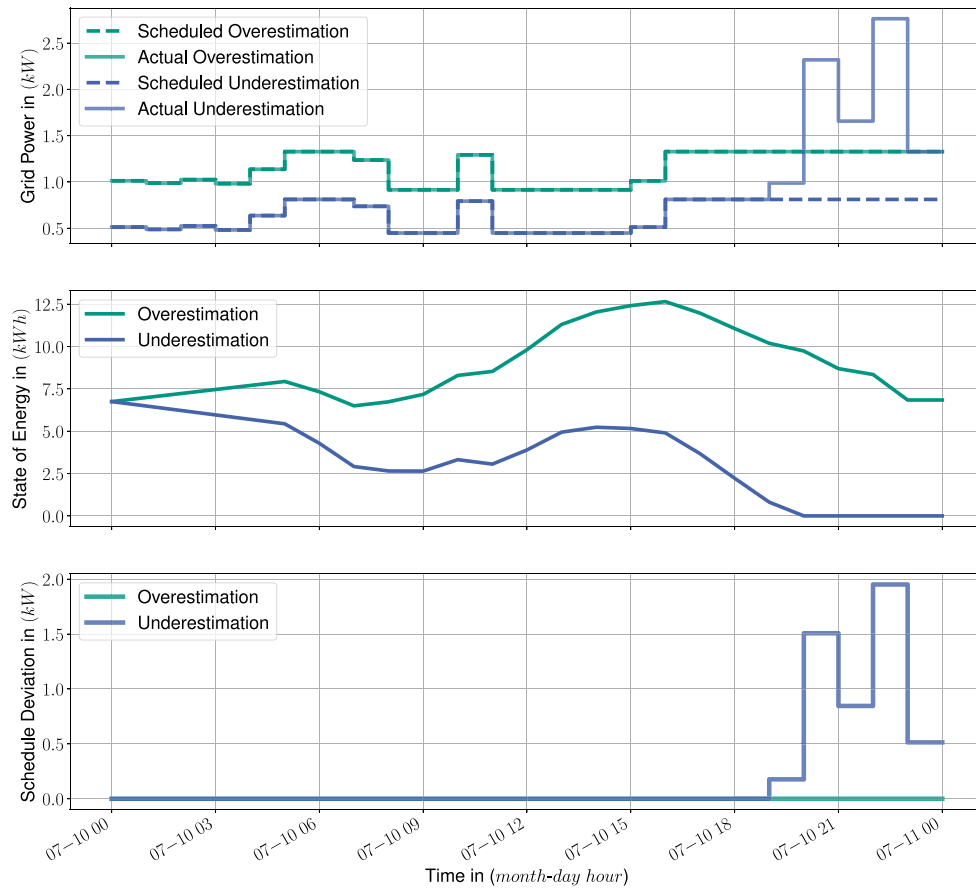
To illustrate the concept of forecast value in a more intuitive way, we present a synthetic toy example. This example highlights the distinction between forecast value and forecast quality by comparing two synthetically generated forecasts. For this purpose, we use load data from Building 109 collected between July 9th at 12:00 PM and July 11th at 06:00 AM, 2012 from the 'Ausgrid - Solar Home Electricity' dataset [49]. The first synthetic forecast consistently overestimates the actual values by 0.25kW across the entire time period. In contrast, the second forecast underestimates the actual values by the same amount. While these forecasts are not realistic in practice, they are intentionally constructed to illustrate a key idea: both forecasts exhibit identical forecast quality in terms of error magnitude. Such similar forecasts, with comparable forecast quality, would naively be expected

¹ The value function $V(F(x, \theta), y)$ returns the forecast value. Since, in our case, the value function is defined as system costs, minimizing the value function implies maximizing the forecast value for the downstream optimization.

² Code available on GitHub: [GitHub-Link](#).



(a) Synthetic forecasts and ground truth data from Building 109 for the period from July 9th to July 11th, 2012.



(b) Resulting dispatch schedule (top), actual state of energy over time (mid), and observed deviations (bottom) with respect to the synthetic forecasts.

Fig. 2. Synthetic example showing the discrepancy of forecast value and quality.

Table 1

Daily Total Costs (rounded to two decimal places), Mean Absolute Error (MAE), and Mean Square Error (MSE) for the synthetic example. Even though both synthetic forecasts perform identical according to quality metrics (MAE/MSE), they affect the optimization problem differently. Consequently, the synthetic forecasts differ in their value for the specific optimization problem at hand.

Forecast	Daily Total Cost (€)	MAE (kW)	MSE ((kW) ²)
Overestimation	9.85	0.25	0.0625
Underestimation	23.71	0.25	0.0625

to correspond to the same forecast value. However, different forecasts can lead to different dispatch schedules, and thus result in different forecast values when evaluated within an optimization framework and against the ground truth. Both forecasts with the ground truth values are shown in Fig. 2(a).

We assume an initial and estimated state of energy of 6.75 kWh at the beginning of the scheduling horizon. The resulting schedule, the state of energy progression throughout the day, and the deviation arising from each forecast are depicted in Fig. 2(b). As can be seen, the schedule derived from the overestimation can be maintained, whereas the one based on underestimation leads to deviations in the evening hours. The reason is an empty battery that leads to the deviation from the schedule, as the flexibility is not maintained anymore.

Forecast quality metrics and the corresponding forecast value, reflected in the resulting total costs, are summarized in Table 1. Since deviations are heavily penalized in this scenario, the underestimated forecast leads to significantly higher daily costs compared to the overestimated one. This demonstrates that conventional forecast quality metrics are not necessarily aligned with the actual performance of the forecast within the optimization context. Consequently, there exists a clear discrepancy between forecast value and forecast quality in the context of the dispatchable feeder problem. A more detailed discussion of forecast quality versus forecast value for dispatchable feeder can be found in [13].

4.3. Used data

The 'Ausgrid - Solar Home Electricity' dataset [49] is used for evaluation. It includes time series data on electricity consumption and solar generation from 300 residential buildings over three years, from July 1, 2010, to June 30, 2013. The data is recorded at 30 min intervals using gross metering. The residential buildings, part of the Ausgrid electricity network and subject to the residential electricity tariff, are selected randomly.

We resample the data to an hourly resolution. The prosumption is calculated by subtracting the PV power generation from the load. Unlike the approach in [15], we do not scale prosumption by a factor of 0.5. Furthermore, we divide the data into training, validation, and test sets using the first year (July 1, 2010, to June 30, 2011) as the training data set, the second as the validation data set (July 1, 2011, to June 30, 2012), and the last year as the test data set (July 1, 2012, to June 30, 2013). Further, we split within the building dimension and use the first 50 buildings for fine-tuning different data usage strategies. In addition, the last 200 buildings were taken for method evaluation. The optimization is performed consecutively starting on July 9, 2012, and ending on June 28, 2013; therefore, covering 355 days. The buildings 51–100 are omitted in evaluation as they are involved in surrogate network training.

4.4. Used models

An overview of the models used is given in Table 2. As an exemplary foundation model, we use the Moirai foundation model [18]. As a replacement for the value function surrogate neural network, we use the same model architecture training configuration as in [15]. In contrast to [15], we adapt the data to fit the data split of training and validation of the test buildings. The Moirai foundation model's large number of total parameters motivates using PEFT methods, see Table 2.

Moirai foundation model. Moirai is a universal forecasting model that demonstrates robust performance across diverse zero-shot scenarios [18]. Serving as an exemplary model within the foundation model class, we leverage the Moirai foundation model [18] in its base configuration to forecast hourly prosumption. Point forecasts are obtained from Moirai by taking the mean from the predictive distribution.

Surrogate neural network. We use an ensemble of five surrogate neural networks, equivalent to [15]. The full architecture can be found in Table B.7. These surrogate networks are generated using the same data and data augmentation techniques described in [15] on the first 50 buildings. The original data from the second 50 buildings are used for validation. We use the same time data split as described in 4.3. An ensemble of five surrogate networks serves as a proxy for $f_{\text{surr}}(F(x, \theta), y, \circ)$, using min-max scaled exogenous data.³ For a more detailed description of the training of the surrogate neural network, see [15], where the same setup is applied.

4.5. Forecasting scenario and fine-tuning setting

An overview of the configurations used to train the PEFT method is shown in Table 3. Using Moirai, we forecast the upcoming 42 h of prosumption, starting at noon. We take the last week of hourly historical prosumption as context without additional features. We take the mean to create a deterministic forecast out of the forecasted distributions by Moirai. Therefore, we configure Moirai using a context length of 168, a forecast horizon of 42, a patch size of 32. For the PEFT adapter, we use an AdamW [50] as an optimizer with a learning rate of 0.0001. We use a batch size of 32 with bfloat16 mixed-precision [51] over 5 epochs.

4.6. Benchmark methods

To evaluate and compare the result of DFF, we use several benchmarks, as described below.

Prediction-focused fine-tuning (PFF). These models align more closely with traditional forecasting methods. Their use to optimize for downstream optimizations suggests that better forecast quality, measured by metrics such as MAE and MSE, is correlated with improved forecast values.

The MAE as loss function refers to the absolute error and minimizes the residuals. It is not sensitive to outliers. Mathematically, it can be written as

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (10)$$

referring to y_i , as the ground truth, $\hat{y}_i$ as the forecast and n as the number of samples.

The MSE metric is more sensitive to outliers, as it weights the errors quadratically. It can be written as

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (11)$$

with also referring to y_i , as the ground truth, $\hat{y}_i$ as the forecast and n as the number of samples.

Zero-shot (ZS). The direct application of the Moirai model without fine-tuning. This represents a traditional zero-shot forecasting approach in the foundation model community.

Naive 48 (N48). We use the last fully observed matching period as a forecast. Thus, the forecasts for the current horizon are based on the values observed 48 h earlier to fit in a full forecast horizon of 42 and adjust it to daily patterns with forecasting at noon. This method assumes that the conditions 48 h ago are similar to the current period.

³ A uniformly drawn state of energy in surrogate training and a simulated state of energy (using MAE forecaster for schedule estimation and MSE forecaster for state of energy estimation) in fine-tuning, and mean, minimum, maximum, standard deviation of the noon hour prosumption.

Table 2
Used models with their respective number of parameters.

Model	Parameters	References
Moirai Base	91 357 728	[18]
5 x Surrogate Neural Network	24753	[15]

Table 3
Moirai configuration and basic fine-tuning settings.

	Setting	Value
Moirai	Patch Size	32
	Context Length	168
	Prediction Length	42
Optimizer	Optimizer	AdamW
	Learning Rate	0.0001
Fine-tuning	Batch Size	32
	Number of Epochs	5
	Precision	bfloat16

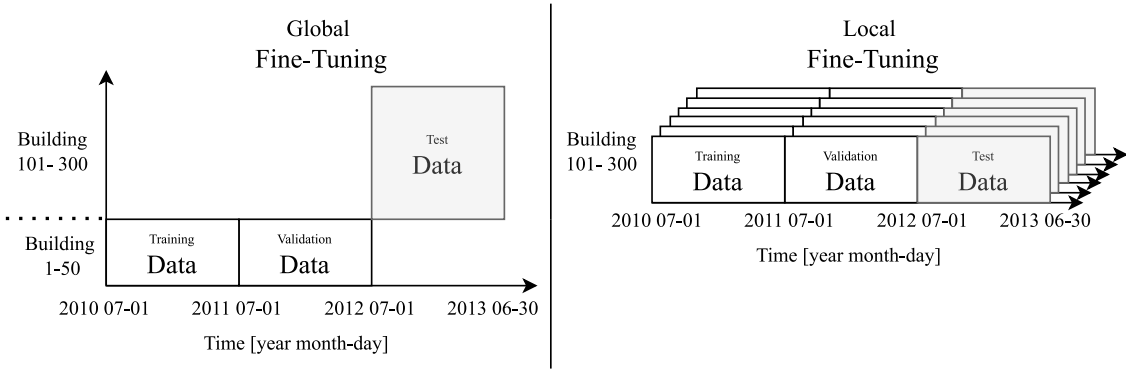


Fig. 3. Visualization of the global and local fine-tuning mode. The global mode performs fine-tuning by considering a set of buildings (1–50), whereas the local mode performs fine-tuning by considering the buildings 101–300 independently. For both methods, the test sets (marked gray) are identical. The buildings 51–100 are omitted in evaluation as they are involved in surrogate network training.

Naive 168 (N168). This method follows the same idea as Naive 48, but it considers weekly periodicity. The forecast is based on values from the same time exactly one week earlier, assuming that last week's observations closely match the current period.

Perfect forecast (P-FC). This method uses the actual observed values as the forecast. While this leads to a perfect forecast with the lowest possible cost, it is an unrealistic assumption because energy load and PV generation cannot be predicted without errors due to their uncertain nature. During operation, no deviations from the schedule occur, and therefore no imbalance costs arise.

5. Impact of fine-tuning modes and PEFT methods in decision-focused fine-tuning

This section introduces the varied settings of the fine-tuning modes and the PEFT methods. Overall, the settings are systematically varied to assess their impact within the DFF method. The first setting relates to the data usage in fine-tuning, and the second explores two versions of PEFT methods, which are introduced in Section 2. The varied settings are also performed using the benchmark methods, if possible, to give a complete overview.

5.1. Comparison of different fine-tuning modes

We compare two fine-tuning paradigms: global fine-tuning and local fine-tuning. Those paradigms focus mainly on the use of data within model fine-tuning. In our case, the building instances used in fine-tuning differ. These concepts are visualized in Fig. 3.

Global fine-tuning. Global fine-tuning refers to fine-tuning a model using data that are not directly related to the specific instance being forecasted. This approach is closely related to zero-shot forecasting, where no customization or fine-tuning is performed for the particular test instance and data. In our case, the model is fine-tuned on data from the first 50 buildings. Strong global model performance suggests the model can generalize well for the optimization task across different instances, even without instance-specific fine-tuning.

Local fine-tuning. In contrast to global fine-tuning, local fine-tuning uses data specific to the forecasted instance, such as an individual building. This approach leverages instance-specific data, aligning the forecasting model with that instance's unique characteristics and patterns and its optimization context. Compared to global learning, this tailored method is expected to enhance performance and suitability instance-specific for downstream optimization, as it directly optimizes for the instance in question.

5.2. Comparison of PEFT methods

The two different configurations used for the PEFT adapter can be seen in Table 4. To compare those two techniques, we keep the basic parameter of LoRA equal to DoRA. Therefore, $r = 8$ is chosen, and $\alpha = 32$. We apply no dropout or additional regularization techniques and only adjust Moirai Network's transformer blocks (v_proj , q_proj , k_proj , and out_proj). This results in DoRA having 626,688 parameters compared to LoRA with 589,824 parameters. Independently of the chosen PEFT method, we only adjust a small number of parameters, i.e., less than 0.7% of the total network parameters.

Table 4

Comparison between LoRA and DoRA. In the top part, the configuration is given in the lower part the number of parameters of the resulting network is shown.

Aspect	LoRA	DoRA
LoRA Configuration	$r = 8$ $\alpha = 32$ $Dropout = 0.0$	$r = 8$ $\alpha = 32$ $Dropout = 0.0$
Target Modules	$v_proj, q_proj,$ k_proj, out_proj	$v_proj, q_proj,$ k_proj, out_proj
Trainable Parameters	589 824	626 688
Total Parameters	91 947 552	91 984 416
Trainable Percentage	0.6415%	0.6813%

Table 5

Configuration Specific Performance Evaluation with Mean and Standard Deviation. Sorted by the Performance within the Forecast Value Average Daily Total Cost. The values are rounded to two decimal places.

Name	Configuration			Performance Factor		MAE (kW)		MSE ((kW) ²)	
	FT. Mode	Loss Func.	PEFT Method	A. D. Total Costs (€)	Std	Mean	Std	Mean	Std
P-FC				4.18					
DFF	Local	Surr.	DoRA	12.75	0.04	0.46	0.00	0.53	0.00
DFF	Local	Surr.	LoRA	12.79	0.03	0.46	0.00	0.53	0.00
DFF	Global	Surr.	LoRA	13.15	0.14	0.57	0.01	0.73	0.03
DFF	Global	Surr.	DoRA	13.18	0.20	0.57	0.01	0.73	0.03
PFF	Local	MSE	LoRA	14.08	0.05	0.38	0.00	0.41	0.00
PFF	Local	MSE	DoRA	14.09	0.02	0.38	0.00	0.41	0.00
PFF	Global	MSE	DoRA	14.13	0.37	0.38	0.01	0.41	0.00
PFF	Global	MSE	LoRA	14.26	0.26	0.38	0.00	0.41	0.00
ZS				15.99		0.42		0.50	
N48				16.77		0.45		0.69	
PFF	Local	MAE	LoRA	17.53	0.04	0.36	0.00	0.43	0.00
PFF	Local	MAE	DoRA	17.57	0.05	0.36	0.00	0.43	0.00
N168				17.83		0.45		0.69	
PFF	Global	MAE	LoRA	18.45	0.26	0.36	0.00	0.43	0.00
PFF	Global	MAE	DoRA	20.37	4.35	0.37	0.03	0.47	0.07

Abbreviations: DFF: Decision-Focused Fine-Tuning; PFF: Prediction-Focused Fine-Tuning; P-FC: Perfect Forecast; ZS: Zero-Shot; N48: Naive 48; Naive N168; A. D. Total Costs: Average Daily Total Costs; PEFT: Parameter Efficient Fine-Tuning; FT: Fine-Tuning; Surr: Surrogate Network; MAE: Mean Absolute Error; MSE: Mean Square Error; DoRA: Directional Rank Adaptation; LoRA: Low-Rank Adaption; Std: Standard Deviation; Func: Function.

6. Results

The presented results are the mean and standard deviation of the forecast value and the forecast quality across five independent runs concerning the mean over the 200 test buildings. [Table 5](#) provides a detailed comparison of various model configurations using three key performance indicators: Average Daily Total Costs (€) as the optimization forecast value, MAE in (kW), and MSE in (kW)² as two forecast quality metrics.

A visualization showing the relation of forecast value and the forecast quality metrics MAE and MSE are shown in [Fig. 4](#). It especially visualizes the relation between the loss and the resulting total costs. It shows further that the choice of the loss function leads to the respective minimization of the performance factor. This visualization is additionally available for different fine-tuning modes and PEFT methods, see [Fig. C.5](#) in [Appendix C](#).

Forecast value. The Perfect Forecast benchmark demonstrates the lowest Average Daily Total Cost, representing an ideal scenario with no discrepancies between forecasted and actual values. With Average Daily Total Costs as low as €4.18, this benchmark allows the system to consistently operate at this minimal cost. While it serves as a valuable

point of comparison, it is essential to note that achieving such perfection is not possible in real-world applications. Regarding Average Daily Total Costs, the DFF configurations trained locally using the surrogate neural network show costs of €12.75 for DoRA and €12.79 for LoRA. In contrast, the costs for the global fine-tuning mode are €13.15 for LoRA and €13.18 for DoRA. The PFF models fine-tuned with the MSE loss function leads to costs of €14.08 for local modes and up to €14.26 for global modes. Models that utilize the MAE loss function have higher costs; the best local model reaches €17.53, while the worst MAE trained model, the global DoRA model reaches €20.37. Notably, PFF fine-tuning using the MAE loss function results in increased total costs compared to the zero-shot usage of Moirai and the Naive 48 benchmark. Furthermore, PFF models trained globally with the MAE loss function perform worse than the Naive 168 benchmark. Comparing the best PFF method (€14.08) with the best DFF (€12.75) method, we observe an improvement of 9.45% in Average Daily Total Costs. If we compare it against the non-fine-tuned foundation model (€15.99) we improve over 20.26% in Average Daily Total Costs.

Forecast quality. For MAE, models fine-tuned with the MAE loss function show values around 0.36 kW up to 0.37 kW for both LoRA and DoRA across local and global fine-tuning modes. Models fine-tuned

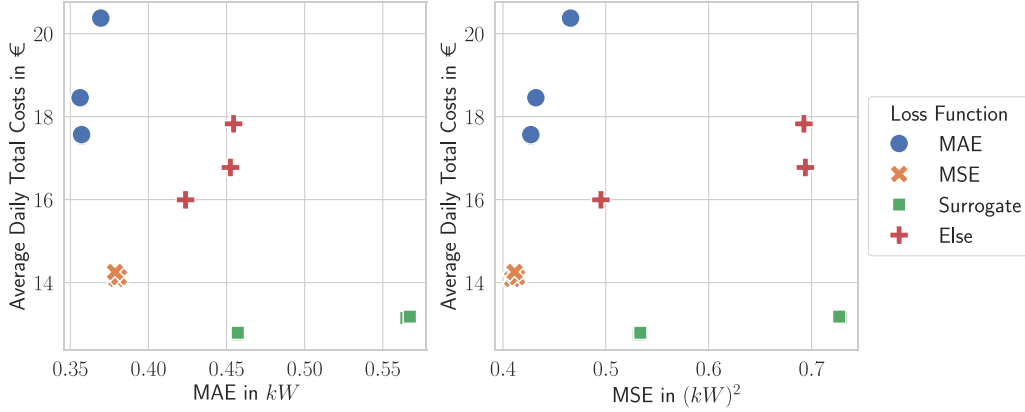


Fig. 4. Average Daily Total Costs created by the different loss functions used versus the forecast quality metrics MSE and MAE. “Else” refers to the Benchmarks with the Perfect Forecast Benchmark being excluded.

with the surrogate neural network show MAE values around 0.46 kW for local fine-tuning modes and 0.57 kW for global fine-tuning modes. When evaluating MSE, models fine-tuned with the MSE loss function show values around 0.41 (kW)², consistent across both local and global modes. Models fine-tuned with the surrogate neural network show MSE values around 0.53 (kW)² for local fine-tuning modes and 0.73 (kW)² for global modes.

Standard deviation. First, it should be noted that the highest standard deviation was observed in the global MAE DoRA configuration, primarily due to an outlier in the global training process. Higher standard deviations are observed in the global setting, both in terms of forecast quality and forecast value. In contrast, the local setting appears to be more stable across the five runs.

7. Discussion

In this section, we discuss four different topics. First, we want to discuss the usage of LoRA compared to DoRA. Afterward, we elaborate on the choice of the loss function, discuss global and local fine-tuning, and conclude on the potential of DFF. We finish this section by pointing out future work.

PEFT methods. DoRA and LoRA show comparable performance in most metrics, with only minor differences. Both methods effectively fine-tune and work across all metrics and their respective loss functions. They are both effective within the DFF methodology. The current study does not provide definitive guidance on choosing DoRA over LoRA. However, regarding parameter efficiency, LoRA demonstrates greater effectiveness because a smaller amount of trainable weights is used, see Table 4.

Loss functions. The choice of loss function affects the model’s performance. This is especially visible in Fig. 4, where the non-linear dependency of the forecast quality and the forecast value is shown. The surrogate neural networks, aligned with the target metric of minimizing Average Daily Total Costs, yield the best results in terms of cost reduction across both fine-tuning modes but also lead to a deterioration in terms of MSE and MAE. MSE and MAE, while useful for minimizing their specific error metrics, lead to higher total costs. MSE is better at managing large deviations, while MAE lowers overall error but incurs the highest costs, especially in global fine-tuning mode. This may result from high imbalance costs, which arise due to outliers within the MAE forecast, which cannot be compensated by flexibility in the battery anymore. Outliers are less prevalent in the MSE forecast, making it more suitable for this optimization problem.

Global vs. Local fine-tuning. Local fine-tuning emphasizes the individual building, improving cost efficiency. However, global fine-tuning offers scalability and generalization across unseen instances, such as buildings, making it more suitable for real-world applications. Although global fine-tuning tends to have higher costs, its ability to generalize across unseen buildings without requiring extensive fine-tuning makes it a more scalable solution for broader deployment, as the training only needs to be done once and does not scale with the number of instances. Thus, depending on the savings and the training costs, local or global fine-tuning might be beneficial. The lower variation observed across the five runs indicates that local fine-tuning exhibits greater stability in applied settings.

Potential of DFF. We observe that fine-tuning within this domain can lead to better decision costs, as we observe an improvement of 20.26% compared to the not fine-tuned model within the zero-shot task. In comparison to PFF models, the improvement also seems promising, with an increase of 9.45%. Furthermore, the concept of DFF can be extended to various domains that rely on predict-then-optimize problems and provide multiple instances that generate data for training surrogate networks. A key requirement is the availability of data that includes ground truth, forecasts, and the resulting forecast value. Potential application areas include energy systems such as demand-side management, peak shaving, electric vehicle charging strategies, and the operation of grid storage systems. Naturally, DFF depends on how the forecast value is defined. Therefore, it is essential to pair its use with meaningful, context-specific value measures. In a demand-side management scenario, this might be the amount of money saved; however, it could also reflect a network perspective, accounting for costs related to grid operation and scheduling. If the costs are asymmetrically or non-linearly coupled to the forecast quality, DFF offers a promising starting point for leveraging time series foundation models in decision-making tasks—shifting the focus beyond purely PFF for optimization.

Future work. Despite the study showing positive results regarding the mean performance across 200 buildings, the averaging process does not examine the impact on individual buildings in detail. Therefore, this approach may be better suited for buildings well-modeled by the surrogate. Since the behavior of each building is not examined in detail, further insights into individual building performance could potentially improve the method.

Additionally, as the approach in [15], this method is not tailored to accommodate a specific convex loss function, meaning the surrogate function does not ensure convexity, which may result in unstable solutions across different buildings. Integrating additional building

variables as exogenous features could improve stability by capturing unique dynamics for each building. Moreover, adapting the model to parameterize a convex loss function might further enhance stability during the fine-tuning process.

Within the DFF method, a comparison of different variants of the PEFT methods and foundation networks or the application of different surrogate neural networks or losses with different properties could be investigated. Further, the current forecasting method does not adhere to exogenous features, this could lead to increased model performance and stabilize the forecaster itself. Further, the potential of global fine-tuning should be evaluated in detail, as it offers a scalable solution across various instances.

Further future work could focus on the applicability and scalability of the approach to other optimization problems that exhibit asymmetric behavior and nonlinear properties. This would enhance its relevance and robustness across the domain of energy-related optimization. Within a specific optimization problem, future work should focus on tailoring the method to capture generalizable patterns across different problem instances of the optimization problem. This could support the development of models toward global applicability and broad usability within a specific optimization problem context.

8. Conclusion

One challenge with integrating optimization problems within the energy systems is the generation of suitable forecasts. This paper begins by identifying three domains: decision-focused learning to generate suitable forecasts, Parameter Efficient Fine-Tuning (PEFT) to efficiently fine-tune models, and time series foundation models as a baseline for fine-tuning. It demonstrates the potential of combining these domains by adapting the decision-focused retraining methodology for fine-tuning of time series foundation models using PEFT. This paper evaluates the Decision-Focused Fine-Tuning (DFF) methodology based on the dispatchable feeder optimization problem, which is a typical battery scheduling optimization problem. The results show that both the Directional Rank Adaptation (DoRA) and Low-Rank Adaption (LoRA) methods are effective with minimal differences in performance. DFF notably enhances model performance, with the surrogate neural network demonstrating a clear advantage in minimizing decision costs. Specifically, DFF outperforms Prediction-Focused Fine-Tuning (PFF) by more than 9.45% in the selected exemplary dispatchable feeder problem. Furthermore, while global DFF leads to more effortless scalability and generalization, local instance-specific DFF remains more cost-effective. Future work should focus on developing scalable strategies to refine the global data strategy, making decision-focused learning methods more applicable to real-world scenarios.

9. Declaration

During the preparation of this work the authors used ChatGPT in order to improve clarity and fluency during the writing process. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRedit authorship contribution statement

Maximilian Beichter: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nils Friederich:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Janik Pinter:** Writing – review & editing, Writing – original draft, Validation. **Dorina Werling:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Kaleb Phipps:** Writing – review & editing, Writing – original

draft, Validation, Software, Methodology, Conceptualization. **Sebastian Beichter:** Writing – review & editing, Writing – original draft, Validation. **Oliver Neumann:** Writing – review & editing, Writing – original draft, Validation. **Ralf Mikut:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Veit Hagenmeyer:** Writing – review & editing, Supervision, Investigation, Funding acquisition. **Benedikt Heidrich:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Veit Hagenmeyer financial support was provided by Helmholtz Association - Energy System Design. Ralf Mikut financial support was provided by Helmholtz Association - Energy System Design. Oliver Neumann financial support was provided by Helmholtz Association - Energy System Design. Dorina Werling financial support was provided by Helmholtz Association - Energy System Design. Sebastian Beichter financial support was provided by Helmholtz Association - Energy System Design. Janik Pinter financial support was provided by Helmholtz Association - Energy System Design. Kaleb Phipps financial support was provided by Helmholtz AI. Maximilian Beichter reports equipment supplied by Helmholtz Association Initiative and Networking Fund. Maximilian Beichter financial support was provided by German Research Foundation (DFG) as part of the Research Training Group 2153 “Energy Status Data: Informatics Methods for its Collection, Analysis and Exploitation”. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The dispatchable feeder optimization problem

A.1. Problem formulation

The mathematical formulation and the parameters are the same as in [15]. The entire operation can be defined as a hierarchical two-level non-convex optimization problem. Within both levels, the time operation is discretized into time intervals indexed by $k \in \mathbb{N}$ with an interval length of $\Delta t \in \mathbb{R}$. The grid interaction is represented solely by the active power exchange with the dispatchable feeder, comprising the first component, the battery power and the second component, the uncertain presumption from the building, summed together. The flexible battery is modeled by its active power input $P_s(k) \in [P_s^-, P_s^+]$ and state of energy $E_s(k) \in [E_s^-, E_s^+]$ with lower and upper bounds $P_s^-, P_s^+ \in \mathbb{R}$ and $E_s^-, E_s^+ \in \mathbb{R}_{\geq 0}$. The progression of the state of energy of the battery is formulated as

$$E_s(k+1) = E_s(k) + \Delta t \cdot (P_s(k) - \mu P_s^+(k) + \mu P_s^-(k)) \quad (\text{A.1})$$

with loss coefficient $0 \leq \mu \leq 1$ and positive and negative directions of the battery’s power input $P_s^+(k) \geq 0$ and $P_s^-(k) \leq 0$.

Based on this description of the two components, the optimization problem’s two levels can be expressed as follows.

First level: Computation of dispatch schedule. The Dispatch Schedule (DS), is calculated within the first level. The DS $\tilde{P}_g(k) \in \mathbb{R}$ is calculated regarding the costs for the following day and considers point forecast of the prosumption $\hat{P}_l(k) \in \mathbb{R}$. The first-level optimization problem is then formulated as

$$\begin{aligned}
 & \min_{\{X\}_{\mathcal{K}}} \sum_{k \in \mathcal{K}} C_{DS}(\tilde{P}_g^+(k), \tilde{P}_g^-(k)) \\
 & \text{s.t. for all } k \in \mathcal{K} \\
 & \quad (A.1) \\
 & \quad E_s(k_0) = E_s^0 \\
 & \quad \tilde{P}_g(k) = P_s(k) + \hat{P}_l(k) \\
 & \quad \tilde{P}_g(k) = \tilde{P}_g^+(k) + \tilde{P}_g^-(k) \\
 & \quad \tilde{P}_g^+(k) \geq 0 \\
 & \quad \tilde{P}_g^-(k) \leq 0 \\
 & \quad P_s(k) = P_s^+(k) + P_s^-(k) \\
 & \quad P_s^+(k) \geq 0 \\
 & \quad P_s^-(k) \leq 0 \\
 & \quad 0 = P_s^+(k) \cdot P_s^-(k) \\
 & \quad \underline{P_s} \leq P_s(k) \leq \overline{P_s} \\
 & \quad \underline{E_s} \leq E_s(k+1) \leq \overline{E_s}
 \end{aligned} \tag{A.2}$$

with a discrete scheduling horizon $\mathcal{K}$, decision vector $X(k) = (\tilde{P}_g(k), \tilde{P}_g^+(k), \tilde{P}_g^-(k), E_s(k+1), P_s(k), P_s^+(k), P_s^-(k))^T$, parameters $E_s^0, \underline{P_s}, \overline{P_s}, \underline{E_s}, \overline{E_s}$, and point forecasts $\hat{P}_l(k)$. Note, it is necessary to know or estimate the state of energy at the beginning of scheduling $k_0 \in \mathbb{N}$.

Second level: Calculation of the actual dispatch. After the computation of the DS, in the second level, the actual dispatch is calculated for every time interval based on the actual prosumption $P_l(k) \in \mathbb{R}$. Within this level, the aim is to minimize the deviation of the corresponding computed DS $\Delta P_g(k) \in \mathbb{R}$ within the technical constraints. In our problem, the second-level optimization problem can be formulated as

$$\begin{aligned}
 & \min_{X(k)} (\Delta P_g(k))^2 \\
 & \quad (A.1) \\
 & \quad E_s(k) = E_s^k \\
 & \quad P_g(k) = P_s(k) + P_l(k) \\
 & \quad P_g(k) = \tilde{P}_g(k) + \Delta P_g(k) \\
 & \quad P_s(k) = P_s^+(k) + P_s^-(k) \\
 & \quad P_s^+(k) \geq 0 \\
 & \quad P_s^-(k) \leq 0 \\
 & \quad 0 = P_s^+(k) \cdot P_s^-(k) \\
 & \quad \underline{P_s} \leq P_s(k) \leq \overline{P_s} \\
 & \quad \underline{E_s} \leq E_s(k+1) \leq \overline{E_s}
 \end{aligned} \tag{A.3}$$

with decision vector $X(k) = (\tilde{P}_g(k), E_s(k+1), P_s(k), P_s^+(k), P_s^-(k))^T$, parameters $\tilde{P}_g(k), P_l(k), E_s^k, P_s, \underline{P_s}, \overline{P_s}, \underline{E_s}, \overline{E_s}$, and actual dispatch $P_g(k) \in \mathbb{R}$. Note, the state of energy in $k \in \mathbb{N}$ is known. During consecutive executions, the next day's schedule is estimated at noon. As the preceding and new schedules overlap at the beginning of the upcoming day, only the latest calculated schedule is applied.

A.2. Parameter specifications of the dispatchable feeder optimization problem

Table A.6 shows the parameter specifications of the dispatchable feeder optimization problem. Note that the parameter α is set to 10 as in [15,48], which results in a high weighting of the imbalance costs and, thus, high costs for deviations from the DS. Therefore, the main aim in terms of cost reduction is to reduce the occurrence of deviations.

Table A.6

Chosen parameter specifications concerning the given optimization problem.

Parameter	Value
Δt	1 h
$\mathcal{K}$	$\{k_s, \dots, k_s + 29\}^a$
c_q^+	$0.05 (\text{€}/(\text{kWh})^2)$
c_l^+	$0.3 (\text{€}/(\text{kWh}))$
c_q^-	$0.05 (\text{€}/(\text{kWh})^2)$
c_l^-	$0.15 (\text{€}/(\text{kWh}))$
P_s	$-5 (\text{kW})$
$\overline{P_s}$	$5 (\text{kW})$
E_s	$0 (\text{kWh})$
$\overline{E_s}$	$13.5 (\text{kWh})$
μ	0.05
E_s^0	day 1: 6.75 (kWh)
α	10 [48]
c_q^d	$0.05 (\text{€}/(\text{kWh})^2)$
c_l^d	$0.3 (\text{€}/(\text{kWh}))$

^a $k_s \in \mathbb{N}$ starts at midnight and the schedule calculation at noon.

Appendix B. Surrogate neural network architecture

Table B.7 provides an overview of the architecture of the surrogate neural network.

Appendix C. Supplementary result plots

See Fig. C.5.

Data availability

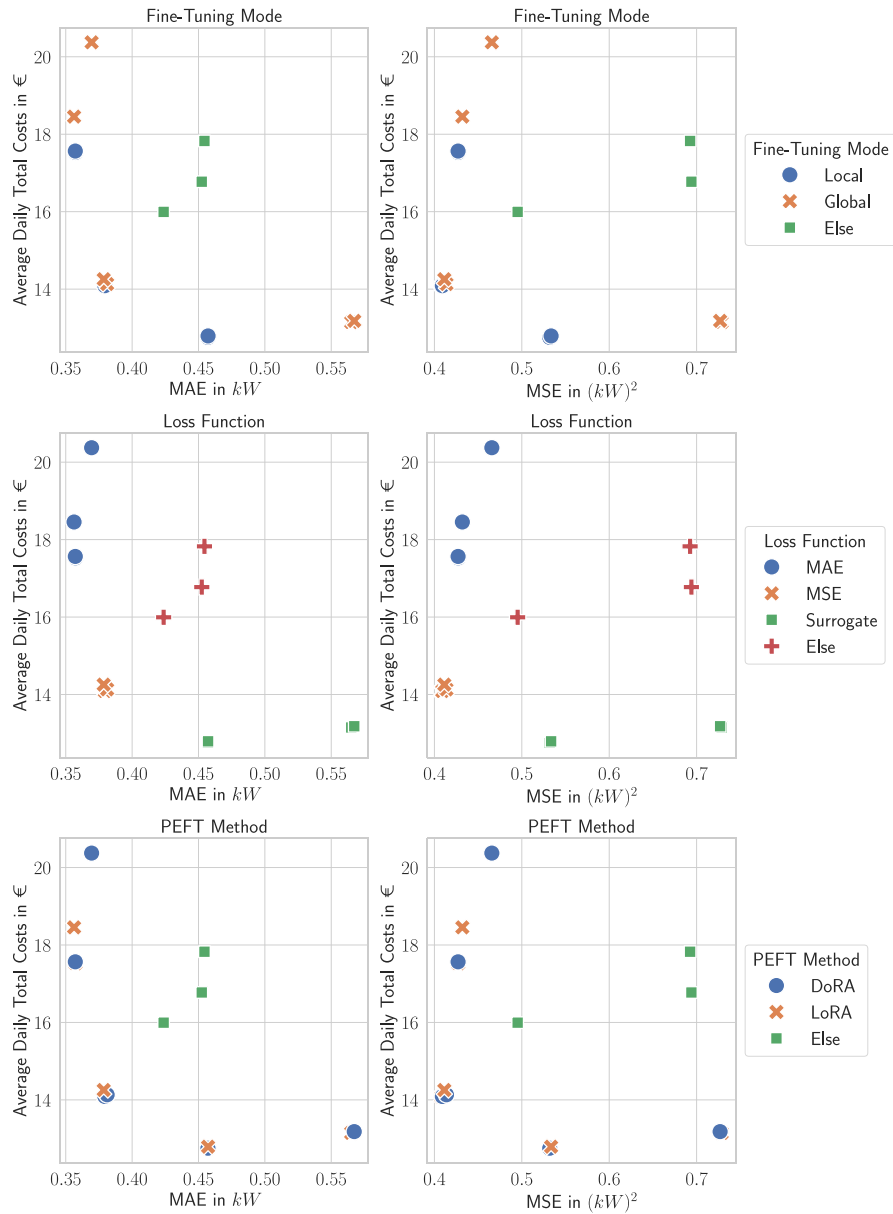
Data will be made available on request.

Table B.7

Detailed overview of the surrogate neural network architecture.

Source: Adapted from [15].

Layer	Input Size	Output Size	Activation	Comments
Split Input in	89	–	–	Whole Input
Split x_0	42	–	–	Forecast
Split x_1	42	–	–	GT
Split x_2	5	–	–	SoE + Stats
FCL0 ₁	42 (x_0)	64	SELU	Forecast Part of Input
FCL0 ₂	42 (x_1)	64	SELU	GT Part of Input
Concatenate	133	133	–	Concatenation (FCL0 ₁ , FCL0 ₂ , x_2)
FCL1	133	64	SELU	–
FCL2	64	64	SELU	–
FCL3	64	64	SELU	–
FCL4	64	32	SELU	–
FCL5	32	8	SELU	–
FCL6	8	1	Sigmoid	Output layer between 0 and 1

**Fig. C.5.** A comparison of forecast value and quality concerning the three factors (Loss Function, Fine-Tuning Mode and PEFT Method). “Else” refers to the Benchmarks with the Perfect Forecast Benchmark being excluded.

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